Chapter 4

Aspects of Particle Adhesion and Removal

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Chapter Outline

4.1 INTRODUCTION

The adhesion of particles to surfaces is of immense technological importance. Moreover, as applications of microminiaturization advance, understanding the factors that control particle adhesion, deposition, and removal becomes more significant. This becomes quite clear when one thinks about such topics as fabrication of microelectromechanical systems (MEMS), uses and properties of buckyballs and nanotubes, and formation of nanoclusters, among others. In some sense, even the definition of “particle” has perhaps changed to include agglomerates of a few atoms or molecules.

The roles of particle adhesion in everyday life range from the quite esoteric to the mundane. For example, we merely have to think of the plethora of products in our own homes that are designed solely for the purpose of removing and collecting dust. Familiar items such as vacuum cleaners and mops, to sophisticated dust collection systems, including electrostatic air cleaning and HEPA filtration, depend on particle adhesion to function properly. New products on the market range from microfiber cloths that take advantage of van der Waals

R. Kohli & K.L. Mittal (Eds): Developments in Surface Contamination and Cleaning, Vol 1, Second edition.
http://dx.doi.org/10.1016/B978-0-323-29960-2.00004-6

interactions to disposable tribocharging cloths that take advantage of electrostatic processes to capture dust without solvents. The expanding knowledge base of particle adhesion is improving our day-to-day quality of life. On the other hand, anyone who suffers from seasonal allergic reactions knows the consequences of the adhesion of pollen particles to mucous membranes.

To illustrate the range of importance of particle adhesion, let us consider a few diverse examples. The most familiar, of course, is the problem of dust in homes. This has spurred a myriad of products and services, some of which were mentioned in the preceding paragraph, including manufacturing and sales of products such as vacuum cleaners and their components (including filters), chemicals to facilitate dusting of furniture, air-duct cleaning services, etc. Add to this list the products sold to remove particulate contamination from dishes and other eating utensils (we have all had to scrub our pots and pans), and one sees that there is a major industry that thrives on controlling particulate contamination in the daily lives of average people.

Particle adhesion has a direct bearing on public health. We are sure that, by this time, most people are aware of the effects of particles of tobacco smoke, asbestos, and coal dust adhering to lung tissue. Indeed, the effects of particulate inclusion and adhesion in biological systems are an active area of research today.[1–3]

One can readily extend the biological connection to particle adhesion by considering cells to be sophisticated particles. Following this path, researchers such as Moss and Anderson[4] and Love and Forsten[5] have related cancer metastasis to particle adhesion. They recognized that metastasis is controlled by a cell detaching from a substrate comprising other cells of the primary cancer site and migrating through the organism until it adheres to a substrate comprising the cells of what will become a secondary site. In related research, Attenborough and Kendall[6] have proposed that the adhesion of erythrocytes is responsible for the formation of blood clots. Other examples of the importance of particle adhesion in the biological and pharmaceutical areas abound.

Particle adhesion has become increasingly important in the fabrication of micro- and nano-devices such as MEMS and microoptoelectromechanical (MOEMS) devices, as well as semiconductor chip fabrication. As the size scale of such devices has decreased, the tolerance for the presence of smaller and smaller particulate contaminants has decreased. Moreover, the ability to clean such surfaces has become more complicated. This is due, in part, to the increasing difficulty encountered in removing such particles without damaging the surface. The reader is referred to the series of books edited by Mittal[7] for a more in-depth discussion of this topic.

There are many other examples of the commercial importance of particle adhesion. Air contamination control has led to the development of electrostatic air and HEPA filters that are commonly used today. To avoid hydrocarbon emissions, automobile manufacturers, among others, are using dry paint particles in place of traditional painting techniques. Grinding and polishing, and the related problem of wear, are associated with the formation of particles due to contact and subsequent separation of surfaces. Sintered part fabrication relies on the ability of particles to cohere strongly enough to avoid fracturing. And the list goes on.

The importance of particle adhesion is not simply a case of devising particles that stick to substrates, such as the example of dry paint, or removing particles from unwanted areas, such as in semiconductor device fabrication. In some instances it is necessary to have particles adhere to and release from the same substrate during a process. An example of this is electrophotography, as commonly practiced with copiers and laser printers.

In the electrophotographic process, an electrostatic latent image is formed on a photoconducting member by first uniformly charging that member and then image-wise exposing it using, for example, a laser scanner. The latent image is developed into a visible image by depositing electrically charged polymer particles comprising a pigment. These particles are commonly referred to as toner. Despite the fact that these particles are highly charged with the same polarity, they must both cohere and adhere to the photoconductor if a usable image is to be produced, throughout the rest of the electrophotographic cycle. The image is next transported through the printer to a place where it is transferred to paper upon application of an electrostatic field. Again, the image must be detached from one substrate and adhered to another while maintaining cohesion between the toner particles. It is important that most of the toner transfers at this point. However, it is also important that the toner should not detach from the photoconductor prior to reaching the point of actual transfer. The image-bearing paper must now be sent to a fuser, where the toner is heated to a temperature above its glass transition temperature T_g and subjected to pressure. This alters the area of contact between toner particles and between the toner particles and the paper, thereby permanently fixing the image. The photoconductor is then cleaned to remove residual toner, generally using a scraper blade or a rotating brush, so that it is ready for the process to start again. It should be clear that cohesion, adhesion, and abhesion (the opposite of adhesion) must be carefully controlled throughout this process if an electrophotographic device is to produce usable images.

This article describes the forces, interactions, and factors that control the adhesion and cohesion of particles. It also discusses how the adhesion of particles to surfaces is commonly determined.

4.2 INTERACTIONS GIVING RISE TO PARTICLE ADHESION

Particles adhere to surfaces because of both the existence of attractive forces between the particles and the substrates, and the mechanical responses of these materials to these attractive forces. Specifically, the interactions between a particle and a contacting substrate cause stresses in both materials. These stresses cause the materials to deform and it is the combination of the deformation and the strength of the interactions that determines how strongly a particle is bonded to a substrate. This and the subsequent sections present the general physics of these interactions, enhancing the engineer's ability to understand and solve technical problems. A more in-depth treatment of particle adhesion is given elsewhere.[8,9]

Forces of nature are often categorized as one of four types: strong interactions, weak interactions, electromagnetic interactions, and gravitational interactions. Which of these are important in a given instance depends on the size of the particles and the separations between the particles or between the particles and the substrates. Consider, for example, the realm of the high-energy physicist. His is a world of baryons and leptons, where the size scales of both the particles and their separation distances are less than 10^{-4} Å. At this range, the dominant forces are generally the strong and weak interactions, although gravity can play a significant role at very short distances such as those encountered when subatomic particles undergo gravitational collapse, as occurs in a black hole. For the size scales of interest in this discussion, these interactions are negligible and will not be considered further.

For the case of macroscopic particles, gravitational forces become quite significant. Let us consider, for example, the case of a spherical particle of radius R and mass density ρ at the surface of the earth. The gravitational force F_g acting on such a particle is given by

$$F_g = \frac{4}{3}\pi\rho R^3 g \tag{4.1}$$

where g is the acceleration due to gravity and equals 9.8 m/s^2. For a particle with a radius of 5 μm and a density $\rho = 1$ g/cm^3, the gravitational force, F_g, is $\sim 5 \times 10^{-12}$ N. While this force is quite small for this particle, its dependence on the cube of the radius implies that this force will increase rapidly with increasing particle size. The effect of gravity is shown in Figure 4.1.

Finally, let us consider the role of electromagnetic interactions in particle adhesion. To do so, however, requires that we further subdivide these interactions into magnetic, electrostatic, and electrodynamic interactions. Furthermore, let us first consider the role of magnetic interactions.

Magnetic interactions are important for certain classes of materials that are of technological importance today, ranging from magnaflux devices used to detect cracks to magnetic recording materials to magnetic carriers used in electrophotographic engines to charge and transport the toner particles. Unfortunately, there has not been much research in the area of the adhesion of magnetic particles and this remains a rich field for enterprising scientists and engineers to explore.

Let us now turn our attention to electrostatic interactions. Electrostatics comprises those areas in which a net transfer of charge has occurred (although not necessarily between the contacting bodies), leaving at least one of the materials possessing an electric charge, and those materials in which there has been a net displacement of charge without a transfer, thereby leaving no

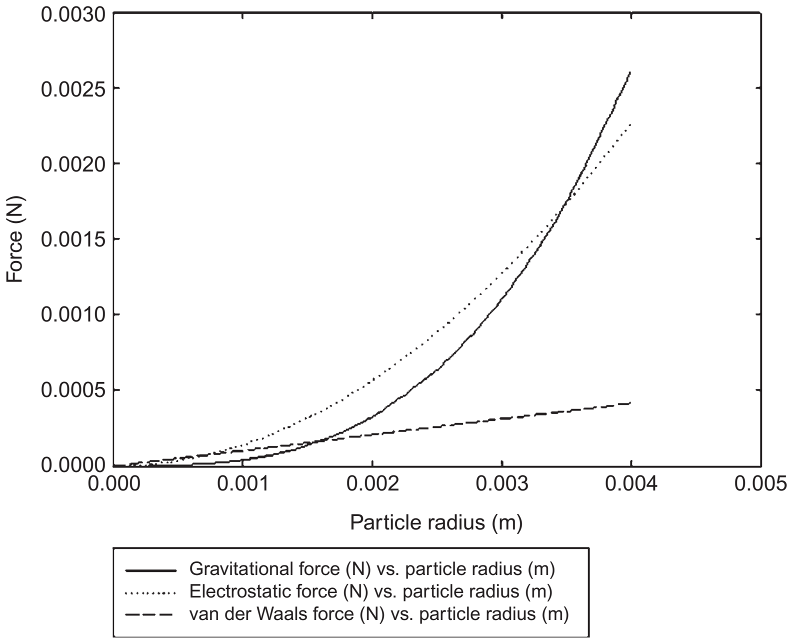

FIGURE 4.1 The gravitational, electrostatic, and van der Waals forces acting on a particle in contact with a substrate. Note how van der Waals interactions dominate for small particles, whereas gravity is the dominant force for large particles and electrostatic forces dominate for intermediate-size particles.

net charge (or monopole) but with a dipole and/or other multipole moment. This set of interactions includes particles that have a net charge through triboelectric interactions,[10] as well as adhesion through surface chemistry[11] be it ionic, covalent,[12] metallic,[13] or Lewis acid-base interactions.[14]

For the most part, the chemical interactions giving rise to particle adhesion are more limited than those giving rise to adhesion in general. The world we live in is dirty. An exposed surface chemically reacts with oxygen or other atoms very rapidly. Moreover, air-borne organics rapidly coat any exposed surface. In short, there are only a relatively few instances, such as lunar dust[15] or nanoclusters formed in ultra-high vacuum on clean substrate surfaces, in which chemical mechanisms play a role in adhesion. Although interactions leading to particle contamination in space may be of practical importance, such a discussion would be divergent from the main themes of this chapter and book and, therefore, should be left to a separate paper. Similarly, a discussion of the adhesion of particles with permanent dipole moments, such as electrets, would be of more limited interest than this chapter would allow and the interested reader is referred elsewhere[16] for a more in-depth discussion. Therefore, the present discussion will focus on the effect of electric charge on the adhesion of particles.

Most particles have an electric charge q. This can be caused through triboelectrification[17–19] or upon the initial formation of the particle through a process known as fractoemission.[20] The presence of a charge on a particle can generate an attractive force to a substrate by polarizing the substrate. This can occur whether the substrate is a conductor or an insulator. As an example, consider the case of a uniformly charged dielectric (insulating) spherical particle of radius R in contact with an electrically conducting, grounded substrate. Because it is conducting and grounded, the potential V must be zero everywhere within the substrate. This can only occur if the substrate is polarized in such a way as to be equivalent to a charge, equal in magnitude but opposite in polarity located a distance, R, within the substrate. The presence of this equivalent charge allows the force of attraction to be calculated using the so-called method of images.[21,22]

Once it is recognized that the charge distribution within the substrate is equivalent to the presence of an image charge, the attractive electrostatic force F_E between the particle and the substrate can be calculated from the Coulomb force law. Accordingly,

$$F_E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{2R}\right)^2 \tag{4.2}$$

where ε_0 is the permittivity of free space, 8.85×10^{-12} $C^2/(N\text{-}m^2)$, and the factor of 2 in the denominator occurs because the separation distance from the center of the particle to the image charge is twice the radius of the particle.

If, as commonly occurs, the charge on the particle is due to triboelectrification, it is plausible to assume that, after multiple contacts, the charge is proportional to the surface area of the particle. Accordingly, under this circumstance,

$$q = 4\pi R^2 \sigma \tag{4.3}$$

where σ is the surface charge density. Upon substituting Equation (4.3) into Equation (4.2), one finds that

$$F_E = \frac{\pi\sigma^2 R^2}{\varepsilon_0} \tag{4.4}$$

The variation of the electrostatic contribution to the detachment force with the square of the particle charge is shown in Figure 4.2. The effect of particle size on the electrostatic force is shown in Figure 4.1. Upon comparing Equations (4.1) and (4.4), it is obvious that, while both F_g and F_E increase with particle radius, F_g does so at a faster rate than does F_E. Conversely, as the particle radius decreases, both forces tend toward zero. However, F_g vanishes at a faster rate than does F_E. This implies that there should be some crossover diameter below which the electrostatic forces dominate and above which the gravitational forces are dominant. The crossover radius $R_{g\text{-}E}$ is given by

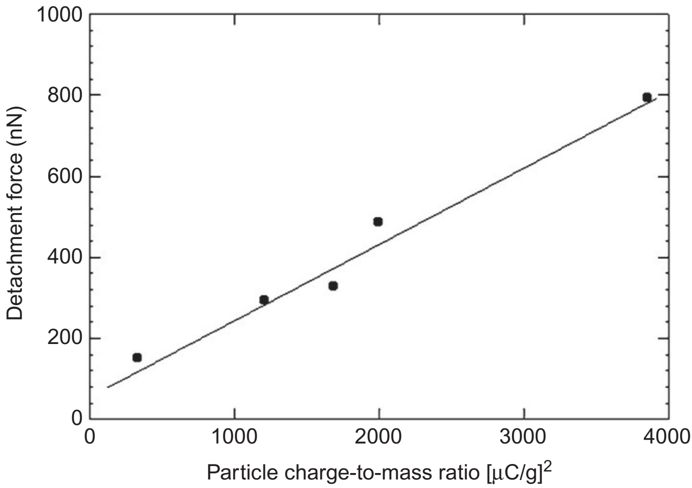

FIGURE 4.2 The effect of electric charge on the detachment force of a spherical polyester particle from an organic photoconductor. The photoconductor had been coated with a monolayer of zinc stearate to reduce van der Waals interactions (from Ref. 23).

$$R_{\text{g-E}} = \frac{3\sigma^2}{4\rho g \varepsilon_0} \tag{4.5}$$

Assuming that $\rho = 10^3$ kg/m^3 and $\sigma = 2 \times 10^{-5}$ C/m^2, which are the values typical of electrophotographic toners, the crossover between F_g and F_E is at ~0.003 m. This point is shown in Figure 4.1.

Whereas there is not much that can be done to alter F_g other than to increase the mass density of the particle or incline the substrate so that only a fraction of the gravitational force is normal to the substrate, F_E can be significantly altered. Most notably, two effects can significantly impact F_E. The first effect is a variation in the surface charge density over the surface of the particle. Such a variation can concentrate charge within a localized region. The presence of such a variation, or charged patch, would bring the image charge closer to the particle, thereby increasing the attractive force. This model has been proposed by Hays.[10] How much such a charge variation can affect the attractive force remains an issue of debate in the literature.[23] Such variations are not expected to account for more than, perhaps, a small increase in the adhesion force.[23] However, this issue is still being investigated.

The second effect arises from the tendency of the dielectric particle to polarize in an electric field, as discussed by Fowlkes and Robinson.[24] Even though the particle in question is electrically insulating, it has a nonzero dielectric constant. Such a constant is the measure of the ability of the positive and negative charges to separate, or become polarized, in the presence of an electric field. For a conductor, the dielectric constant is infinite, as no field can exist within the

LIVE GRAPH
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conductor. The dielectric constant for most insulators (materials such as TiO_2 and $BaTiO_3$ are notable exceptions) is $\sim$3. In effect, the charge on the particle induces an image charge within the substrate. The image charge then polarizes the particle, inducing a dipole. Each pole of the dipole then induces its own image charge within the substrate. These image charges now induce a quadrupole within the substrate. In effect, one gets an infinite, slowly converging series of multipole attractive interactions.

As of this time, there are no experimental data supporting the Fowlkes and Robinson model, but that should not be construed as saying that the model is necessarily incorrect. The model does, however, have several limitations. First, it is assumed that a Gaussian surface could be constructed around the particle. This is not the case for irregular particles or particles that have deformed as a result of adhesion-induced stresses. Moreover, the multipole contributions to the attractive force all depend on the difference between the dielectric constants of the particle and the substrate. However, for most materials, this difference is close to zero, suggesting that polarization effects may not be significant. This would be different, of course, if the particles were close to, but detached from, the substrate. In that case, because of the intervening medium (air), the attractive force between the detached particle and substrate would be greater than that between the attached particle and the substrate. One could readily imagine a scenario in which sufficient force is applied to the particle to detach it. Immediately upon detachment, the attractive force undergoes a stepwise increase, upon which the particle again attaches to the substrate. However, since the detachment force is still present, the particle immediately detaches. This process would repeat ad infinitum. Obviously, further work is needed to validate the multipole model.

Finally, the reader should be reminded that the present discussion has been, for the most part, limited to the electrostatic attraction between a dielectric particle and a conducting substrate. The problem of a dielectric particle interacting with a dielectric substrate has not yet been solved. However, insofar as contact charging[25] and polarization can cause electrostatic forces between the particle and substrate, electrostatic forces do play a role here too.

Let us now focus on electrodynamic forces. These differ from electrostatic forces as there is a time-varying component to the fields giving rise to these forces. Of the set of electrodynamic forces, adhesion arises principally from three distinct interactions that, taken together, are termed "van der Waals forces".[8,26]

In the simplest model, there are three distinct types of interactions that are considered to contribute to van der Waals forces. The first occurs between freely rotating permanent dipoles (Keesom interactions) aligning. The second occurs when a freely rotating permanent dipole induces and aligns with an instantaneous dipole in a neighboring molecule (Debye interactions). The third arises when an instantaneous dipole forms in one molecule and induces an instantaneous dipole in a neighboring molecule (London or dispersion forces). In each case, the interaction potential falls as the inverse of the separation distance between the molecules to the sixth power. Of these three types of interactions, the dominant contribution to van der Waals interactions generally arises from dispersion forces. This is partially due to energetics concerns, as discussed in Refs. 8 and 26, and partially due to the fact that, in solids, the dipoles are constrained and are not freely rotating.

Upon integrating the interaction potentials over all molecules, one could determine the net interaction potential between a spherical particle and a planar substrate. Then by integrating over the distance needed to bring the particle from infinity into contact with the substrate, the force of attraction can be determined. This is discussed in more detail elsewhere.[8] Upon performing such integrations, the force of attraction due to van der Waals interactions F_{vdW} is calculated as

$$F_{\mathrm{vdW}} = \frac{AR}{6z_0^2} \tag{4.6}$$

where z_0 refers to the separation distance between the surface of the particle and the surface of the substrate, and is typically of the order of 4 Å. This is based on the interatomic spacings of van der Waals bonded solids such as argon. The Hamaker coefficient,[27] A, is a measure of the amount of van der Waals interaction between two dissimilar materials or within a pure material and is typically of the order of 10^{-19} to 10^{-20} J.[26] It is system specific, meaning that A is a function of the interacting surfaces and the media surrounding the surfaces. Methods of predicting A using Hamaker's pairwise additive approach[27] or the Lifshitz theory[28] are well established,[26,29] often only requiring the dielectric constant and a small set of indices of refraction of a material. See Ref. 26 for Hamaker coefficients of common systems related to particle adhesion. Using A and z_0, the dependence of the van der Waals forces on the size of the particle is shown by the straight line in Figure 4.1. It should be noted that, in contrast to either the electrostatic or the gravitational forces, van der Waals forces are relatively short range, decreasing as the square of the inverse of the separation distance. In practical terms, this means that, once a particle is detached from a surface by more than 20 to 30 nm, the van der Waals interactions can be ignored.

As with both gravitational and electrostatic forces, van der Waals forces increase with the radius of the particle. However, F_{vdW} increases only linearly with the particle radius. Conversely, all three forces tend toward zero with decreasing particle radius, but van der Waals forces vanish at a slower rate than do either electrostatic or gravitational forces. As was the case with the other two types of forces, there is a crossover radius $R_{\mathrm{vdW\text{-}E}}$, below which the van der Waals forces should be the dominant force and above which the electrostatic or the gravitational forces should dominate. $R_{\mathrm{vdW\text{-}E}}$ can be readily calculated by equating Equations (4.4) and (4.6). Upon doing so, one finds

$$R_{\mathrm{vdW\text{-}E}} = \frac{A\varepsilon_0}{6\pi (z_0\sigma)^2} \tag{4.7}$$

Assuming that $A = 10^{-19}$ J, $\sigma = 2 \times 10^{-5}$ C/m^2, $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2/(N-m^2), and $z_0 = 4 \times 10^{-10}$ m, one finds that $R_{\text{vdW-E}} = 0.0007$ m. However, just as the electrostatic forces can vary, depending on specific charge distribution and dielectric constant, so can the van der Waals attraction. For example, consider the case of a metallic substrate or particle.

As discussed, the largest contribution to van der Waals interactions comes from London forces. As these depend on the polarizability of the materials, one would expect metals to have a higher surface energy than semiconductors, and semiconductors to have higher surface energies than electrical insulators. Indeed, as discussed by Krupp,[30] this is the case. Experiments by Pashley and Tabor[31] showed that the adhesion is very high between a tungsten tip and a nickel substrate prepared and cleaned in an ultra-high vacuum. However, metals rapidly lower their surface energy by forming oxide layers, which should reduce adhesion due to the reduced polarizability of the oxide layer compared to the clean metal surface. In the same paper, Pashley and Tabor also reported that as little as two monolayers of oxide on the nickel surface reduced adhesion by a factor of 3. Oxidation is not the only method by which adhesion can be altered. Any adsorbed species on the surface of a substrate, such as organics, can also reduce adhesion by lowering the substrate surface energy.

Yet another factor that can influence the van der Waals contribution to adhesion is surface roughness. The presence of a few asperities on the surface of the particle or substrate can effectively separate the bulk of the particle from the substrate. Under this scenario, the radius of the particle would no longer control the force of adhesion. Rather, the force of adhesion would depend on the radius of the asperities times the number of asperities in contact with the substrate. This approach is still too simplistic because not all asperities are the same size, nor do they touch with the same pressure. As discussed by Fuller and Tabor,[32] the contacts between asperities and a smooth substrate are broken sequentially, with the smallest asperity contacts breaking first. Essentially, the asperity contacts unzip sequentially from the surface. As a result, there is a factor by which the force needed to detach a single particle from a substrate is divided in order to properly represent the detachment force for a multiple contact system. Measurements by Schaefer et al.[33] suggest that the net detachment force can vary from one-third to one-fifteenth of what one would expect simply based on breaking the bonds to all asperities simultaneously, depending on the system of interest.

Particle roughness is commonly introduced to toner particles by coating the particles with silica clusters having diameters of $\sim$50 nm. As shown in Figure 4.3, the force needed to detach 8.6 μm diameter toner particles from an organic photoconductor decreases monotonically with silica concentration.[34] Along with this study, a number of other studies have been performed to demonstrate the effects that surface roughness have on adhesion.[35–38]

Finally, one might think that asperities can both reduce and increase adhesion, depending on the size and frequency of the asperities on both the substrate

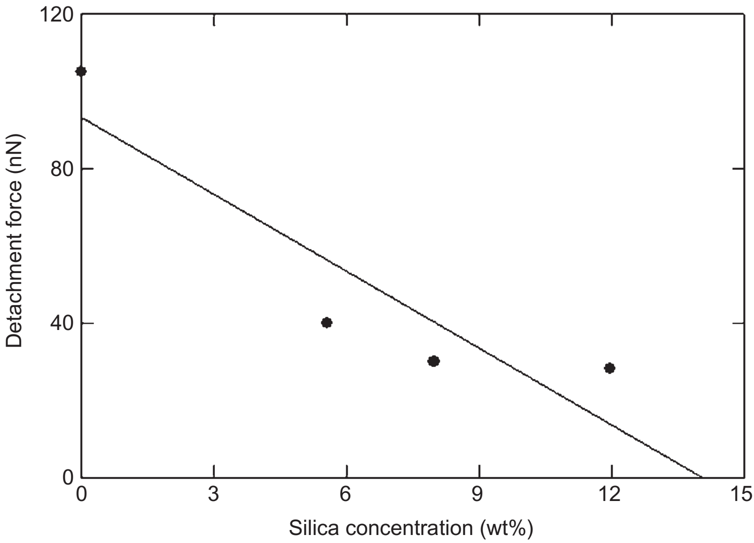

FIGURE 4.3 The effect of the concentration of silica on the surface of 8.6-μm-diameter toner particles on the force needed to detach these particles from an organic photoconductor. The silica particles serve as asperities, thereby creating a roughened surface on the toner (from Ref. 34).

and the particle. The argument for reducing adhesion has just been presented. However, it is conceivable that, if the asperities on both bodies can nestle together, adhesion would actually be increased. In a practical sense, however, this is highly unlikely. The asperities would have to perfectly mesh together to get an increase in adhesion. This would require that the amplitude and periodicity of the asperities on both bodies be exactly equal, as even small divergences would cause a mismatch. This does not normally occur. As a result, the presence of asperities usually decreases the force needed to detach a particle from a substrate.

4.3 MECHANICS OF PARTICLE ADHESION

Thus far, the types of interactions that give rise to particle adhesion have been presented. However, as mentioned earlier, both the interactions and the mechanical responses of the materials to the stresses generated by these interactions determine how strongly a particle is bonded to a surface. Accordingly, we will now turn our attention to discussing the adhesion theories that relate the interactions to the mechanical properties of the materials.

The present-day understanding of particle adhesion is based on the theory proposed by Johnson et al.[39] This theory, generally referred to as the JKR theory, is based on small elastic deformation contact mechanics. It does not allow for any long-range interactions. It assumes that the total energy of the system, which must be at a minimum at equilibrium, is composed of three distinct terms. The first term is the mechanically stored energy, which is simply generated by applying a load P to the particle, thereby creating a displacement of the centers of the contacting materials. The second term is the elastically stored energy that is generated when the materials are deformed. The final contribution to the total energy U_S comes from the surface energy and is given by

$$U_S = w_A \pi a^2 \tag{4.8}$$

where a represents the radius of contact and w_A is the thermodynamic work of adhesion, and is related to the surface energies γ_P and γ_S of the particle and substrate respectively, and their interfacial energy $\gamma_{P\text{-}S}$ by

$$w_A = \gamma_P + \gamma_S - \gamma_{P\text{-}S} \tag{4.9}$$

As discussed in Refs. 8 and 26, the work of adhesion can be directly related to the van der Waals interactions.

After some manipulation, it is found that

$$a^3 = \frac{R}{K}\left\{P + 3w_A\pi R + \left[6w_A\pi RP + (3w_A\pi R)^2\right]^{1/2}\right. \tag{4.10}$$

where

$$K = \frac{4}{3\pi(k_P + k_S)} \tag{4.11}$$

and

$$k_{P(S)} = \frac{1 - v_{P(S)}^2}{\pi E_{P(S)}} \tag{4.12}$$

where the letters P and S denote the particle and substrate, respectively, v refers to the Poisson ratio, and E denotes Young's modulus.

Equation (4.10) is generally referred to as the JKR equation. Although presented in terms of a particle adhering to a flat substrate, it is equally applicable to interparticle cohesion simply by replacing the particle radius with a reduced radius.

In the absence of surface forces, such as those arising from van der Waals interactions, $w_A = 0$ and Equation (4.10) reduces to

$$a^3 = \frac{RP}{K} \tag{4.13}$$

which the reader may recognize as the Hertz equation for an elastic indenter. On the other hand, in the absence of any applied force, Equation (4.10) reduces to

$$a = \left(\frac{6\pi w_A}{K}\right)^{1/3} R^{2/3} \tag{4.14}$$

Equation (4.14) implies that there is a finite contact between the particle and the substrate simply due to adhesion.

The JKR theory also allows the force needed to separate a particle from the substrate to be determined. Specifically, Equation (4.10) predicts the actual contact radius as a function of the applied and adhesion forces. As such, the solutions to Equation (4.10) must be real. Now, consider the case where one is attempting to detach a particle. This is accomplished by applying a sufficiently large negative force to the particle. However, it is important that the radicand in Equation (4.10) does not become negative, as this would predict a complex, rather than real, contact radius, which is a physical impossibility. As a result, detachment occurs when a force P_{detach} is applied such that

$$P_{\mathrm{detach}} = -\frac{3}{2} w_{\mathrm{A}} \pi R \tag{4.15}$$

It should also be noted that according to the JKR theory, it is not necessary for the contact radius to vanish as P approaches P_{detach}. Rather, detachment occurs at a finite contact radius that is ~63% of the equilibrium contact radius in the absence of an external load. Finally, it should be noted that according to the JKR theory, the force needed to detach the particle from the substrate is independent of the Young modulus of either material.[40]

The lack of dependence on elastic-plastic properties assumed by the JKR theory does make some sense, as all interacting materials are assumed to behave elastically.[26,41] However, real systems often exhibit some plastic behavior, leading to viscoelastic behavior and adhesion hysteresis.[42] Further, real systems are not atomically smooth. The asperities present on the interacting surfaces will therefore deform more freely than an atomically smooth surface. This is due to the "free space" surrounding the asperity into which the material may deform. Deformation into a flat surface may experience a resistive force by the bulk material. The deformation of the asperity increases the effective area of contact, which results in an increased force required for detachment. As Young's modulus decreases, or as the propensity for the asperity to deform increases, the detachment force would naturally increase. This effect has been confirmed experimentally by Gady et al.[40]

As previously discussed, the JKR model only allows for forces to be present within the actual area of contact. Because of the short range of van der Waals forces, this approximation is valid when the dominant mode of interactions arises from surface forces. However, contributions to particle adhesion caused by electrostatics can be significant.

Let us now consider the case where the particle is attracted and attached to the substrate by both electrostatic and van der Waals forces. The electrostatic forces are clearly long-range and, therefore, cannot be simply assumed to vanish once contact between the particle and substrate has been broken. Rather, a formal treatment of this problem would require that the fundamental treatment of particle adhesion by the JKR theory be reexamined, with the energy required to bring a charged particle from infinity to the substrate added to the other three contributions. Tang et al.[43] show that the JKR theory can be used to accurately predict the contact radius of a uniformly charged particle to a rigid half-space by properly complementing the mechanical force with an electrostatic force. The contributions to particle adhesion resulting from van der Waals and electrostatic interactions can be separated out by measuring the detachment force as a function of the square of the particle charge, as shown by Rimai and coworkers.[23,44] The y-axis intercept, corresponding to $q=0$, approximates the van der Waals contribution to the force, and the difference between the measured force and the van der Waals contribution gives the electrostatic component.

An alternative theory of particle adhesion was proposed by Derjaguin et al.[45] and is generally referred to as the DMT theory. This theory, rather than being energy-based as is the JKR model, uses a force balance approach and assumes that the shape of the contact region is Hertzian. As a result of the differing assumptions, for a given size contact radius, the DMT theory predicts a work of adhesion that is approximately six times greater and a removal force that is approximately one-third greater than that predicted by the JKR model.[46] This discrepancy was apparently resolved by Muller et al.,[47] who showed that the JKR model was valid for systems comprising lower Young's moduli and larger particles, whereas the DMT model was valid for more rigid systems and smaller particles. It is the view of the present authors that, in practical terms, the JKR model reasonably describes particle adhesion for just about any size particle if one of the materials is elastomeric. Moreover, the present authors further believe that, for most cases, one need not be concerned with the DMT model, which would be expected to be valid for submicrometer particles on harder substrates. This is because, for such materials, the deformations would generally be plastic rather than elastic, as will be discussed presently.

That adhesion forces between a particle and a substrate could induce plastic flow was first proposed by Krupp.[30] Subsequently, Maugis and Pollock (MP)[48] extended the JKR theory to allow for total plasticity. They found that the contact radius was related to the yield strength Y of the lower-strength material by

$$P + 2\pi w_{\mathrm{A}} R = 3\pi a^2 Y \tag{4.16}$$

In the absence of any applied load, Equation (4.16) reduces to

$$a = \left(\frac{2w_{\mathrm{A}}R}{3Y}\right)^{1/2} \tag{4.17}$$

The difference in the power-law dependence of the contact radius on particle radius for the JKR and MP theories gives an easy method of distinguishing elastic and plastic deformations.

If plastic deformations are to occur, the adhesion-induced stresses must exceed Y. As discussed elsewhere,[8] the average surface force-induced pressure p_{m} between a spherical particle and a substrate is given by

$$p_m = \frac{2w_A}{z_0} \tag{4.18}$$

For most materials, the calculated pressures are comparable in magnitude to Young's modulus, implying that adhesion-induced strains are quite large. In fact, certain materials such as elastomers would be capable of supporting such strains without yielding. More rigid materials should yield at such large strains. Indeed, Bowen et al.[49] reported that the adhesion-induced deformations between micrometer-size soda-lime glass particles and a silicon substrate were plastic rather than elastic.

Calculation of the detachment force in the case of plastic deformations is not as straightforward as it is for elastically deformed materials. Certainly, negative values for P can be substituted directly into Equation (4.16). As there are no cases that would result in complex solutions for a, as is the case with the JKR theory, one could simply infer that detachment occurs when a vanishes. This would be correct when the deformations remain plastic even as the size of the negative load is increased, corresponding to decreasing strains in the materials. This is often referred to as ductile fracture. However, as the magnitude of P increases, upon application of a negative load, the total stress decreases. Often, the stress can decrease to the point where the deformations are elastic. The detachment force would then have to be determined by the JKR or DMT criteria, depending on which theory is applicable under the particular circumstances.

The act of deforming plastically is not instantaneous. Rather, over time, the size of the contact area increases under load until some equilibrium is established. As the size of the contact increases, the tenacity with which the particle is bonded to the substrate should also increase. This effect was observed by Krishnan et al.,[50] who determined the contact radius and detachment force as a function of time for submicrometer polystyrene latex spheres on a silicon substrate. They found that the adhesion-induced contact area increased logarithmically with time. Corresponding to that increase, the force needed to separate the particle from the substrate also increased logarithmically, with the same time constant. These results suggest that if one wishes to remove particles from a surface, one should clean that surface as soon after the particles are deposited as possible.

Ultimately, the challenge of examining the effect of plastic deformation on particle adhesion and the related detachment force is an area of ongoing study. Maugis and others have been publishing in this area for several decades. Notably, the Maugis–Pollock theory is based on the existing models for elastic contact (Hertz, JKR, DMT, etc.), but compensates for plastic deformation.[48,51] Further studies have been conducted by numerous others, specifically to evaluate the adhesion of particles in contact under plastic deformation effects.[41,42,52–70] Abbott and Firestone[71] and Greenwood and Tripp[54] have looked exclusively at fully plastic contact. Walton and Braun[66] expanded plastic effects to unloading. Many have considered the elastic-plastic contact effects on adhesion.[52,53,60,63,68,70] Some have employed finite element modeling to evaluate adhesion where plastic deformation may occur.[41,42] A thorough briefing of these models and their contributions to the study of plastically deforming particles in contact is discussed in Ref. 72.

4.4 FACTORS AFFECTING PARTICLE ADHESION

As previously discussed, the two dominant forces that affect adhesion arise from electrostatic and van der Waals interactions. van der Waals interactions, being relatively short range, can be reduced by adding asperities to particles, as is commonly done by coating the surface of electrophotographic toners with silica.[73] Other agents that coat the substrate, such as zinc stearate or Teflon,[74] also reduce the detachment forces. Representative data showing the effects of such coatings on the detachment force of polystyrene particles from a polyester substrate are shown in Figure 4.4. While there is some debate as to the mechanisms by which these occur, there is evidence that these materials reduce the van der Waals interactions without altering the electrostatics.[23] Obviously, altering one or both sets of interactions by physically changing the properties of the materials would alter the radius at which a given interaction is dominant. Similarly, the distance from a substrate is important. Electrostatic air filters work because of the long-range nature of the electrostatic forces. Relative differences in the position of the contacting materials in the triboelectric series, along with the number of times the two materials contact each other, also affect the detachment force.[75]

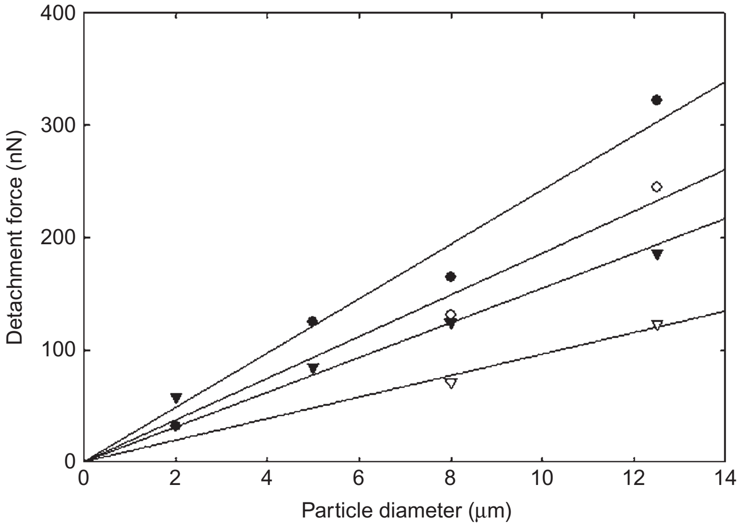

FIGURE 4.4 The force needed to detach polystyrene spheres, as a function of particle diameter, from a bare (solid circles) polyester substrate, as well as one that had been coated with a thin layer of silicone (open circles), Teflon (solid triangles), and zinc stearate (open triangles) (from Ref. 74).

Finally, some discussion about the effects of relative humidity (RH) on particle adhesion is in order. Water can affect particle adhesion in two ways. First, it can alter the surface resistivity, thereby altering surface charge distributions and possibly affecting the formation of multipoles.[76] Moreover, there could be capillary condensation between a hydrophilic particle and substrate which could cause a capillary force. This force typically increases with increasing RH and often dominates over van der Waals forces.[77]

Analytical models describing the relationship between RH and the capillary force are well established for smooth, ideal particles and surfaces, as reviewed by Butt and Kappl.[78] Experimentally, however, the adhesion force between hydrophilic surfaces can monotonically decrease or increase, or exhibit a maximum with increasing RH, while hydrophobic surfaces are generally independent of RH.[79–83] Thus for non-ideal, experimental surfaces, hydrophilicity merely allows for the presence of a capillary force, but its dependence on RH is heavily influenced by other factors.[84] For example, capillary forces decrease with increasing surface roughness[82,85–89] (provided that the two surfaces do not complement each other); sharper tips/asperities tend toward maxima in the capillary force, whereas duller tips/asperities tend toward monotonic increases in capillary force as RH increases[79,90]; and micrometer-sized particles generally do not exhibit a capillary force below 50% RH, but nanoscale tips and asperities do show significant dependence below 50% RH.[88,91] Other factors such as surface heterogeneity,[78,92] local temperature fluctuations,[93] kinetics,[91,94] and non-equilibrated systems[95] also affect these trends.

4.5 METHODS OF MEASURING THE ADHESION OF PARTICLES TO SUBSTRATES

This chapter would not be complete without some discussion of the techniques used to determine the adhesion of particles to substrates, though a detailed discussion is beyond the scope of this chapter. The reader is referred to Ref. 8 for a more complete discussion of this topic.

It is helpful to define what is meant by the term "particle adhesion". Specifically, it is necessary to distinguish between the attractive forces that bring a particle to a surface in the first place and the magnitude of the force needed to detach the particle from the surface. Both have applications and are important, but they are not always equal. Because of the mechanical responses of the materials to stress, the detachment force must be determined from energetics considerations, rather than by simply assuming that detachment occurs when the forces of attraction are balanced by the applied forces. With this in mind, let us first look at some techniques that allow one to measure the detachment force.

One commonly used technique to measure the force needed to separate a particle from a substrate is centrifugation.[30,96–99] Knowledge of applied centrifugal forces is used to characterize adhesion forces through the centrifuge technique. Detachment of a particle adhered to a surface that is perpendicular to the centripetal force will occur when the adhesion force is exceeded by the centripetal force. It is important to remember that this only holds true if the centripetal force is acting perpendicular to the surface on which the particle is adhered. A modified centrifuge tube is needed to accomplish this by holding the substrate parallel to the axis of rotation. This technique works well for particles having radii between ~3 and 100 μm, depending on both the particle and substrate properties.

To carry out the technique, particles are deposited onto a substrate and the number of particles on that substrate is determined initially. The centrifuge is then operated at an initial speed for the first run, with sequentially higher speeds after each run. Upon completion of each run, the surface is removed and the number of particles remaining adhered is determined as a function of the rotational speed of each run. Data using this method are represented by a graph of the percentage of particles still adhering to the surface against the adhesion force or rotational velocity. Typically, the geometric median adhesion force is assumed to represent the mean adhesion force for the system. This characteristic adhesion force for the centrifuge method is defined as the force where 50% of particles are still adhered to the surface.[100]

An important aspect of the centrifuge method is the capability of measuring the adhesion forces for a collection of particles. This provides better representation of the adhesion forces for the whole particulate system compared to single-particle adhesion measurements. The method also makes it possible to measure the adhesion of particles on different surfaces or the same surface with varying surface roughnesses in a single experiment.[101]

Determination of the detachment force via the application of an electrostatic field is another commonly used technique. In general, the particles are deposited on an electrode that forms part of a parallel-plate capacitor having a controlled spacing. This method has several advantages and disadvantages compared to centrifugation. The size range of particles that can be evaluated with this technique is comparable to that of centrifugation. Good statistics can be obtained. In some instances, electrostatic detachment provides a closer simulation to the process of interest than does centrifugation. An example of this is the electrophotographic transfer process. Measurements can be made in shorter periods of time, in some cases, than is possible with centrifugation.[76] In addition, measurements can be made in a variety of atmospheres. There are four principal disadvantages to this technique. First, only electrically conducting substrates can be used. Second, the forces that can be applied are often lower than in centrifugation. Third, the application of an electric field can interact with the adhesion of the particle by polarizing the particle. Fourth, the charge on the particle must be known and well controlled.

Busnaina et al.[102] have pioneered a technique that allows the detachment force of particles in the tenths of a micrometer diameter range to be determined. This technique involves rotating the substrate bearing the particles while impinging a liquid stream. Using standard equations from hydrodynamics, the detachment force can be determined. The principal advantage of this technique is that it allows detachment forces to be determined for particles that are too small for centrifugation or electrostatic techniques. It also allows good statistics to be obtained. Its principal disadvantage is that it introduces a liquid, which, in turn, can alter the attractive forces.

Colloidal probe microscopy (CPM) is also a common technique for measuring both attractive and detachment forces of individual particles to surfaces.[33,88,91,103–107] In CPM, a particle (usually of the order of micrometers in diameter) is attached to an atomic force microscope (AFM) cantilever. Then, a piezoelectric column slowly raises the substrate into contact with the attached particle through an applied voltage. As the surface gets close to the particle, adhesion forces dominate and the particle adheres to the surface causing downward deflection of the cantilever, known as the snap-in. The piezoelectric column continues to extend, going past the initial plane of the cantilever, causing upward deflection of the cantilever. With the particle still adhered to the surface, the piezoelectric column is retracted causing the cantilever to deflect downwards again. The retraction continues to the point where there is enough force to break the adhesion of the particle to the surface. This is referred to as the pull-off force. A laser directed to the tip of the cantilever is reflected through a mirror to a photodetector to detect displacement of the cantilever. The tracking of displacement provides a force-distance curve of the snap-in, deflection, and pull-off. It is assumed that the cantilever deflects according to Hookean mechanics. Thus, the spring constant of the cantilever, determined from experiment, multiplied by the displacement of the pull-off force, results in a measured detachment force.

Though adhesion and detachment forces may be measured with piconewton accuracy, CPM can only determine the detachment force of a single particle at a time. This greatly limits the ability to use statistical techniques in comparison to centrifugation. The orientation of the particle is also fixed, so the point of the particle contacting the substrate may not be representative of the entire particle.[87] On the other hand, this technique allows known loads that increase the deformation to be applied to the particle.[108] It can be performed in a variety of atmospheres, including vacuum.[109] The time of contact can be varied from seconds to minutes, allowing some time variation, and also allowing exploration of short-time effects.[89,110,111] It also allows a precise mapping of the adhesion properties of the substrate and allows the local adhesion properties to be correlated precisely with particle and substrate roughness, Young's modulus variations, and other particle characteristics. Finally, this technique allows the contribution of triboelectric interactions to be directly determined. See Refs. 111 and 112 for a more complete description of CPM and its utility.

DeMejo et al.[113] did not measure the detachment force directly. Rather, they determined the adhesion-induced contact radius as a function of particle radius.

From those data, they were able to determine the work of adhesion and then infer a detachment force using Equation (4.15). The principal advantage of this technique is that the particle and substrate are in equilibrium. This is not the case for methods where the particle is actually removed. On the other hand, one may be interested in a dynamic removal force that simulates a specific process.

Finally, Gady et al.[75] developed several AFM-based methods that allow the attractive force between a particle and a substrate to be determined. Gady et al. determined the attractive force between the particle and the substrate as a function of displacement by measuring the deflection of the AFM cantilever at separation distances greater than that where the jump-to-contact occurs. They also determined the force gradient as a function of displacement by measuring the resonance frequency of the cantilever. Using these independent measurements, they were able to determine both the electrostatic and van der Waals contributions to the attractive force at a given separation.

4.6 SUMMARY AND CONCLUSIONS

Much understanding of particle adhesion has been gained over the past few decades. This knowledge has increased our understanding of the science behind adhesion, in general, and allows engineers to more readily control processes that involve particle adhesion phenomena. There are, however, a number of questions that need to be answered.

Understanding the factors that contribute to the adhesion of particles to surfaces is extremely important in areas such as cleaning and contamination control, with the specific means employed to control contamination and clean surfaces depending on the application. For example, in semiconductor fabrication, relatively few particles having diameters less than 1 μm and adhering to the surface of silicon wafers by van der Waals forces can adversely affect the production of integrated circuits. In this instance, the size of the contaminants would preclude the use of devices such as blade cleaners or brushes. However, the number of devices fabricated per unit time is sufficiently low as to allow the jetting of CO_2 snow particles to clean the surface.

At the other extreme of removing particles from a surface is the waiter using a blade to remove crumbs from a dinner table prior to the patrons ordering coffee and dessert. In this example, the particles are sufficiently large so that gravitational forces are the dominant interaction. Here, the waiter uses the blade to sweep the particles into a small dustpan or similar device.

Dusting furniture or cleaning electronic devices or electrical equipment are examples where van der Waals, electrostatic, and gravitational forces are all significant. The effect of gravitational forces is evident by the fact that dust predominantly settles on the top surface of furniture. On the other hand, if that piece of furniture were to be turned upside down, the dust would not fall off. Here, damp dust cloths, so-called "tack cloths" comprising an agent that makes the cloth sticky, various sprays and liquids are commonly used to reduce electrostatic forces and increase the adhesion of the particle to the cloth to effect cleaning. Various sprays, waxes, and other fluids are also used to reduce the adhesion to the furniture and electronic devices.

Electrophotographic devices such as laser printers, especially those operating in commercial establishments, also have demanding cleaning requirements. Toner particles are typically 8 μm in diameter and are highly charged. They are attracted to an image-wise electrically charged photoreceptor during a process called "development". After development, the image is transferred to paper by pressing the paper against the photoreceptor and applying an electric field to attract the toner particles to the paper. In an electrophotographic engine, residual toner left on the photoreceptor after transfer as well as paper fibers and paper filler, such as submicrometer particles of calcium carbonate, must be removed. While more particles can be left on the photoreceptor after cleaning in electrophotographic engines compared to semiconductor device fabrication tools, a far greater number of particles must be removed at a much higher process speed and cover a much larger area than that encountered in the semiconductor industry. Here, van der Waals and electrostatic forces are dominant and cleaning often employs corona chargers to control the electrostatic forces, electrically conducting fur brushes, blades, and similar devices.

It should be apparent from the above discussion that the choice of the means of cleaning must consider the particle-substrate interactions as well as the process speeds.

Particle adhesion has also become an increasingly important subject in seemingly unrelated phenomena ranging from the mechanisms by which pressure-sensitive adhesives work to how individual cancers metastasize.

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